Session 5: Sampling Techniques & Sampling Bias

Population vs. Sample, Random Sampling Simulation & Bias Analysis

1. Population vs. Sample Examples (App Context)

Three real-world scenarios contrasting entire populations with representative samples:

App Context	Population (Complete Set)	Sample (Subset)
Instagram	All 1,000,000+ followers of a top sports brand account worldwide.	A subset of 1,000 randomly selected followers polled for feedback.
Zomato	Every restaurant active on the platform across all cities in India.	500 randomly chosen restaurants in Bengaluru to estimate average prep time.
Flipkart	All 50,000+ smartphone products listed on the entire platform.	100 top-reviewed smartphones selected to analyze price-to-rating trends.

2. Python Random Sampling Simulation

Task: Simulate random selection of 50 users out of 1,000 using Python's `random.sample()`.

```
import random # Generate a population of 1,000 fake user IDs population = [f"user_{i}" for i in range(1, 1001)] # Set seed for reproducible sampling random.seed(42) # Select a random sample of 50 unique users survey_sample = random.sample(population, 50) print(f"Total Population Size: {len(population)}") print(f"Selected Survey Sample ({len(survey_sample)} users):") print(survey_sample[:10], "... [truncated for display]")
```

SAMPLED RESULT OUTPUT

Total Population Size: 1000
Survey Sample (50 IDs): ['user_654', 'user_112', 'user_7", 'user_781', 'user_280', 'user_633', 'user_301', 'user_18', 'user_389', 'user_222', ...]

3. Why Swiggy Uses Sampling Instead of Full Data

Practicality, Speed & Resource Efficiency

Analyzing every single order ever made on Swiggy involves hundreds of millions of historical database records, making query execution computationally expensive, slow, and resource-intensive. Using a statistically sound random sample yields near-identical insights and confidence intervals in seconds, drastically reducing computational overhead while enabling rapid real-time decision-making.

4. Sampling Bias Analysis (Spotify Scenario)

Scenario: A general survey on music preferences is distributed solely to Spotify users who recently streamed Bollywood songs.

IDENTIFIED BIAS: SAMPLING BIAS / SELECTION BIAS (SPECIFICALLY SAMPLING FRAME BIAS)

Explanation: This sampling method systematically excludes listeners of other genres (such as Classical, Hip-Hop, or EDM) who do not consume Bollywood music. Because the survey audience is pre-filtered by genre, the sample fails to represent the overall Spotify user base, leading to heavily distorted results that overrepresent Bollywood preference across the entire platform.

5. Sampling Strategies for BookMyShow

Demonstrating two distinct probability sampling methods to study movie booking habits among BookMyShow users:

1. Stratified Sampling

Application: Divide BookMyShow users into non-overlapping groups (strata) based on language preference (e.g., Hindi, English, Tamil, Telugu). Next, draw a random sample proportional to the size of each group.

✓ **Ensures minority language moviegoers are proportionally represented without subgroup omission.**

2. Systematic Sampling

Application: Arrange the chronological database of all ticket purchases. Choose a random starting index (e.g., user #14), and then select every *k*-th user (e.g., every 50th user) who completes a movie booking.

✓ **Simple to automate in production pipelines and guarantees balanced coverage across all booking hours.**

SUMMARY COMPARISON

While **Stratified Sampling** controls for demographic representation across distinct categories, **Systematic Sampling** offers an efficient algorithmic method to continuously sample transactional stream data.